

Maximum a posteriori (MAP) estimation

Computational Biology II

28th April, 2026

Likelihood and log-likelihood

- Previously, we introduced the likelihood function:

$$L(\theta) = p(D \mid \theta) = \prod_{i=1}^n p(d_i \mid \theta)$$

where θ is the set of model parameters, $p(\cdot)$ is a probability density function, and $D = \{d_1, \dots, d_n\}$ are i.i.d. observations.

Likelihood and log-likelihood

- Previously, we introduced the likelihood function:

$$L(\theta) = p(D \mid \theta) = \prod_{i=1}^n p(d_i \mid \theta)$$

where θ is the set of model parameters, $p(\cdot)$ is a probability density function, and $D = \{d_1, \dots, d_n\}$ are i.i.d. observations.

- Maximum likelihood estimation (MLE):

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

Likelihood and log-likelihood

- Previously, we introduced the likelihood function:

$$L(\theta) = p(D \mid \theta) = \prod_{i=1}^n p(d_i \mid \theta)$$

where θ is the set of model parameters, $p(\cdot)$ is a probability density function, and $D = \{d_1, \dots, d_n\}$ are i.i.d. observations.

- Maximum likelihood estimation (MLE):

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

- Maximizing likelihood is equivalent to maximizing log-likelihood, given as

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log p(d_i \mid \theta)$$

Posterior distribution of parameters

- So far, we focused on $p(D \mid \theta)$. What about $p(\theta \mid D)$?

Posterior distribution of parameters

- So far, we focused on $p(D | \theta)$. What about $p(\theta | D)$?
- **Bayes theorem** gives us

$$p(\theta | D) = \frac{p(D | \theta)p(\theta)}{p(D)}$$

where:

- $p(\theta | D)$ is the **posterior**.
- $p(D | \theta) = L(\theta)$ is the **likelihood**.
- $p(\theta)$ is the **prior**.
- $p(D)$ is the **evidence term**.

Maximum a posteriori (MAP)

- Instead of maximizing likelihood (MLE), we may decide to maximize posterior (MAP):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta \mid D)$$

Maximum a posteriori (MAP)

- Instead of maximizing likelihood (MLE), we may decide to maximize posterior (MAP):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta \mid D)$$

- Again, it will be useful to maximize its logarithm instead:

$$\begin{aligned} \log p(\theta \mid D) &= \log \left(\frac{p(D \mid \theta)p(\theta)}{p(D)} \right) \\ &= \log p(D \mid \theta) + \log p(\theta) - \log p(D) \end{aligned}$$

Maximum a posteriori (MAP)

- Instead of maximizing likelihood (MLE), we may decide to maximize posterior (MAP):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta \mid D)$$

- Again, it will be useful to maximize its logarithm instead:

$$\begin{aligned} \log p(\theta \mid D) &= \log \left(\frac{p(D \mid \theta)p(\theta)}{p(D)} \right) \\ &= \log p(D \mid \theta) + \log p(\theta) - \log p(D) \end{aligned}$$

- From that, we get

$$\hat{\theta}_{MAP} = \arg \max_{\theta} \ell(\theta) + \log p(\theta)$$

Example: ridge regression

- Consider a linear model parametrized by $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$

Example: ridge regression

- Consider a linear model parametrized by $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$
- We assume Gaussian prior on the parameters:

$$\beta_i \sim N(0, \sigma^2) \quad \forall i \in \{1, \dots, p\}$$

where the density function is

$$p(\beta_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta_i^2}{2\sigma^2}\right)$$

Example: ridge regression

- Consider a linear model parametrized by $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$
- We assume Gaussian prior on the parameters:

$$\beta_i \sim N(0, \sigma^2) \quad \forall i \in \{1, \dots, p\}$$

where the density function is

$$p(\beta_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta_i^2}{2\sigma^2}\right)$$

- We have

$$\log p(\beta_i) = \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{\beta_i^2}{2\sigma^2}$$

Example: ridge regression

- Consider a linear model parametrized by $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$
- We assume Gaussian prior on the parameters:

$$\beta_i \sim N(0, \sigma^2) \quad \forall i \in \{1, \dots, p\}$$

where the density function is

$$p(\beta_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta_i^2}{2\sigma^2}\right)$$

- We have

$$\log p(\beta_i) = \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{\beta_i^2}{2\sigma^2}$$

- MAP leads to the penalty:

$$\lambda \sum_{i=1}^p \beta_i^2, \quad \lambda = \frac{1}{2\sigma^2}$$